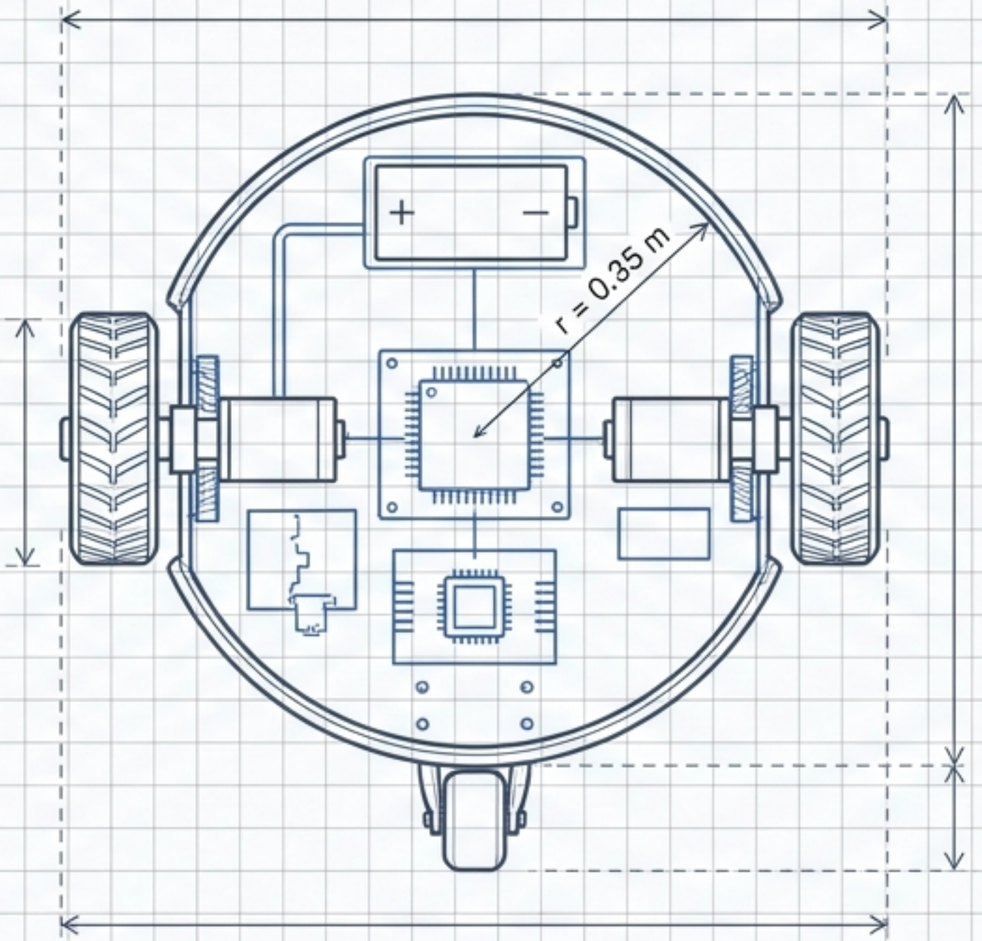
# FaultShield: Cross-Layer Reliability Benchmark & Recovery Agent

Bridging the gap between physical sensor faults and cloud AI latency for autonomous mobile robots.

NVIDIA Nemotron

Nebius Token Factory

Zero Physical Hardware




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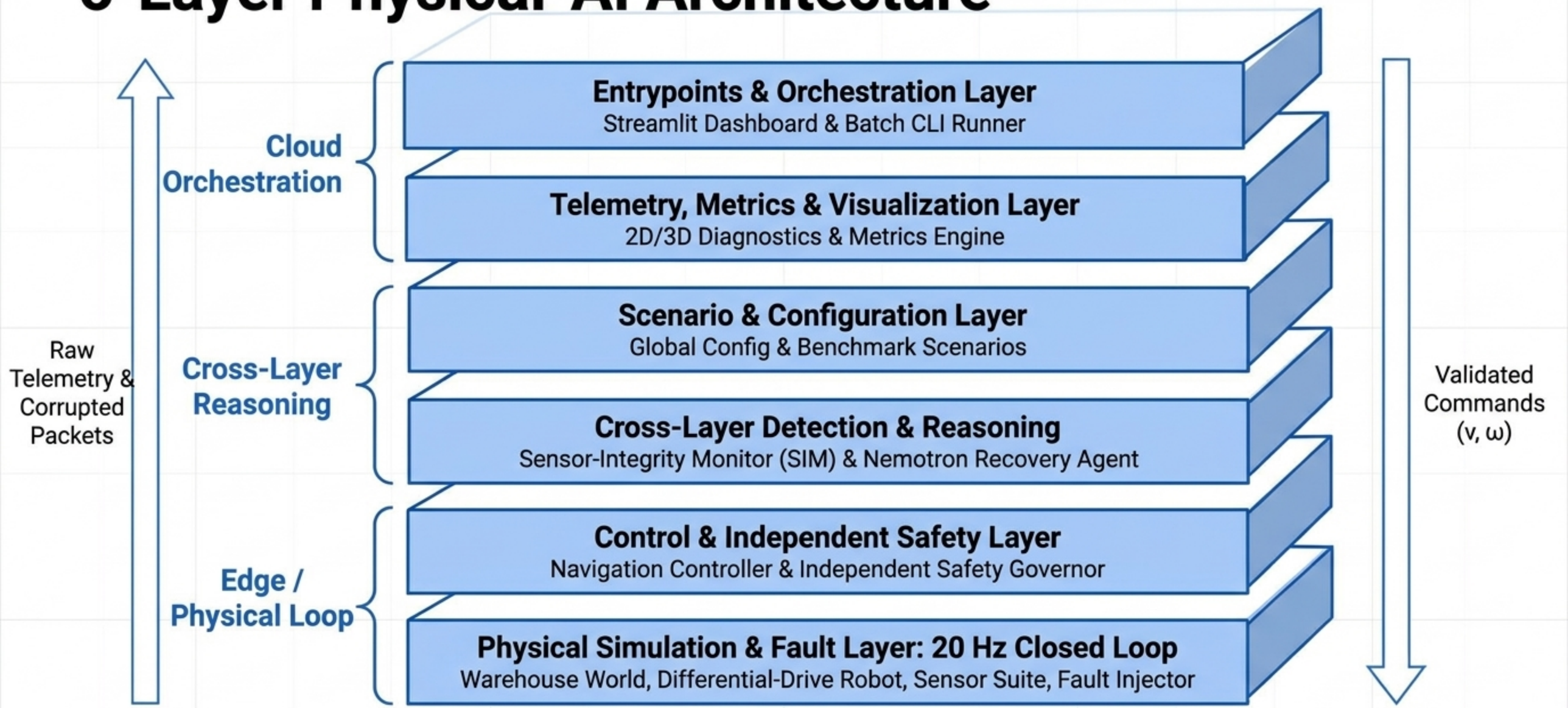
# The Unresolved Cross-Layer Gap

| Prior Art | Scope Addressed | Unaddressed Cross-Layer Gap |
|---|---|---|
| VLA-FixBench (ICML 2026) | VLM diagnosis of tabletop manipulation failures | Does not evaluate mobile robot kinematics under corrupted/frozen range & IMU telemetry |
| LIBERO-Recover (2026) | Task-level policy retries and behavioral adaptation | Assumes accurate, uncorrupted state estimation and zero sensor freeze |
| LARA (2026) | Asynchronous inference delay compensation | Does not couple remote LLM inference delay with simultaneous onboard sensor faults |
| Industrial Patents | High-level anomaly reporting & generic uncertainty | Lacks a deterministic Time-To-Live (TTL) Safety Governor paired with allowlisted LLM recovery |

**The Core Question:** What happens when corrupted physical data meets cloud network latency? Can the agent detect when not to act?

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# 6-Layer Physical-AI Architecture



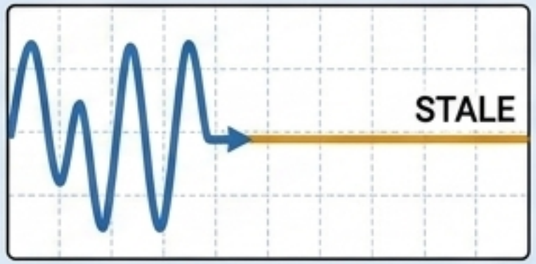

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# Cross-Layer Fault Injection Engine

Simulating physical sensor failures without mutating ground-truth physics.

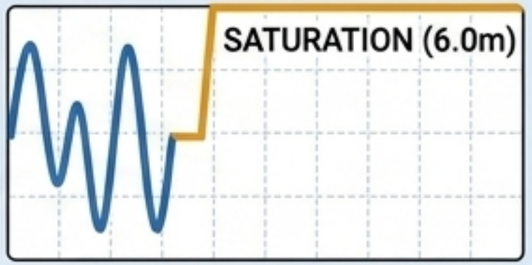
## LIDAR_FROZEN

Locks LiDAR buffer to the last healthy scan frame (stale timestamp).



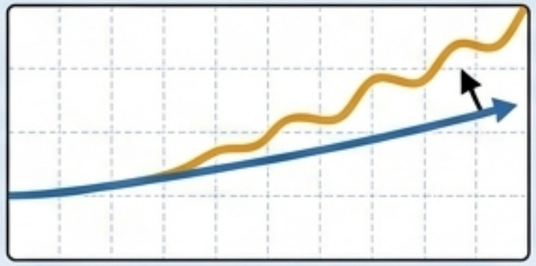

## LIDAR_DROPOUT

Forces all 24 beams to saturate at max range (6.0m), blinding the controller.



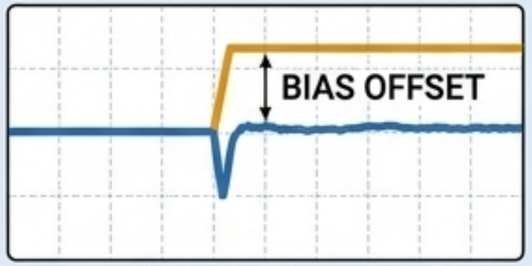

## ODOMETRY_DRIFT

Injects cumulative heading drift (0.25 × mag rad/s) and linear velocity bias.



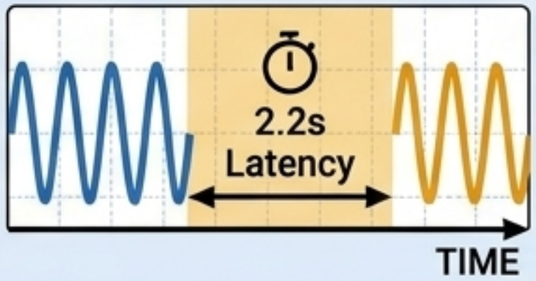
## IMU_BIAS

Injects acute gyroscope yaw-rate bias (+0.55 × mag rad/s).



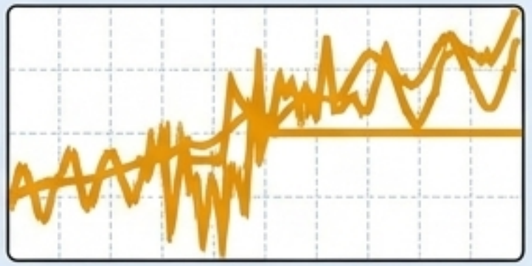

## AI_DECISION_DELAY

Couples sensor dropout with a configurable cloud reasoning delay (e.g., 2.2s).



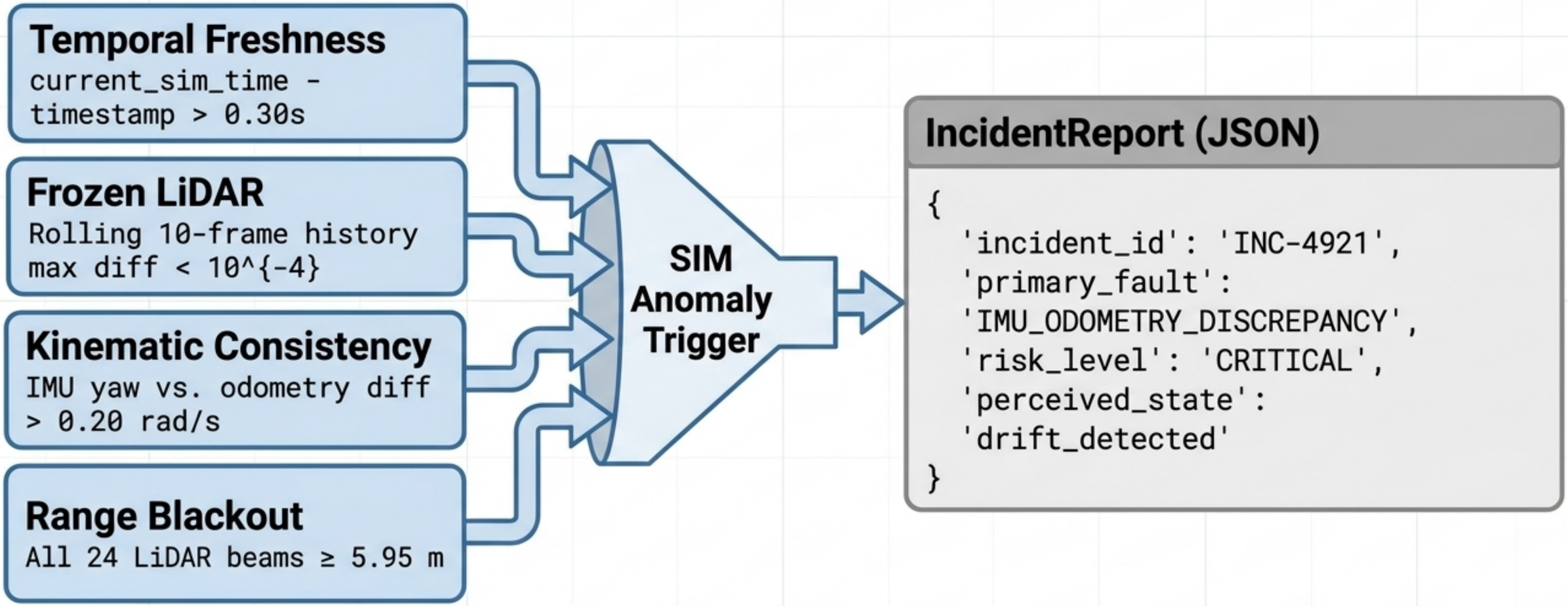

## COMPOUND

Simultaneous Frozen LiDAR + Odom Drift + IMU Bias + Inference Latency.



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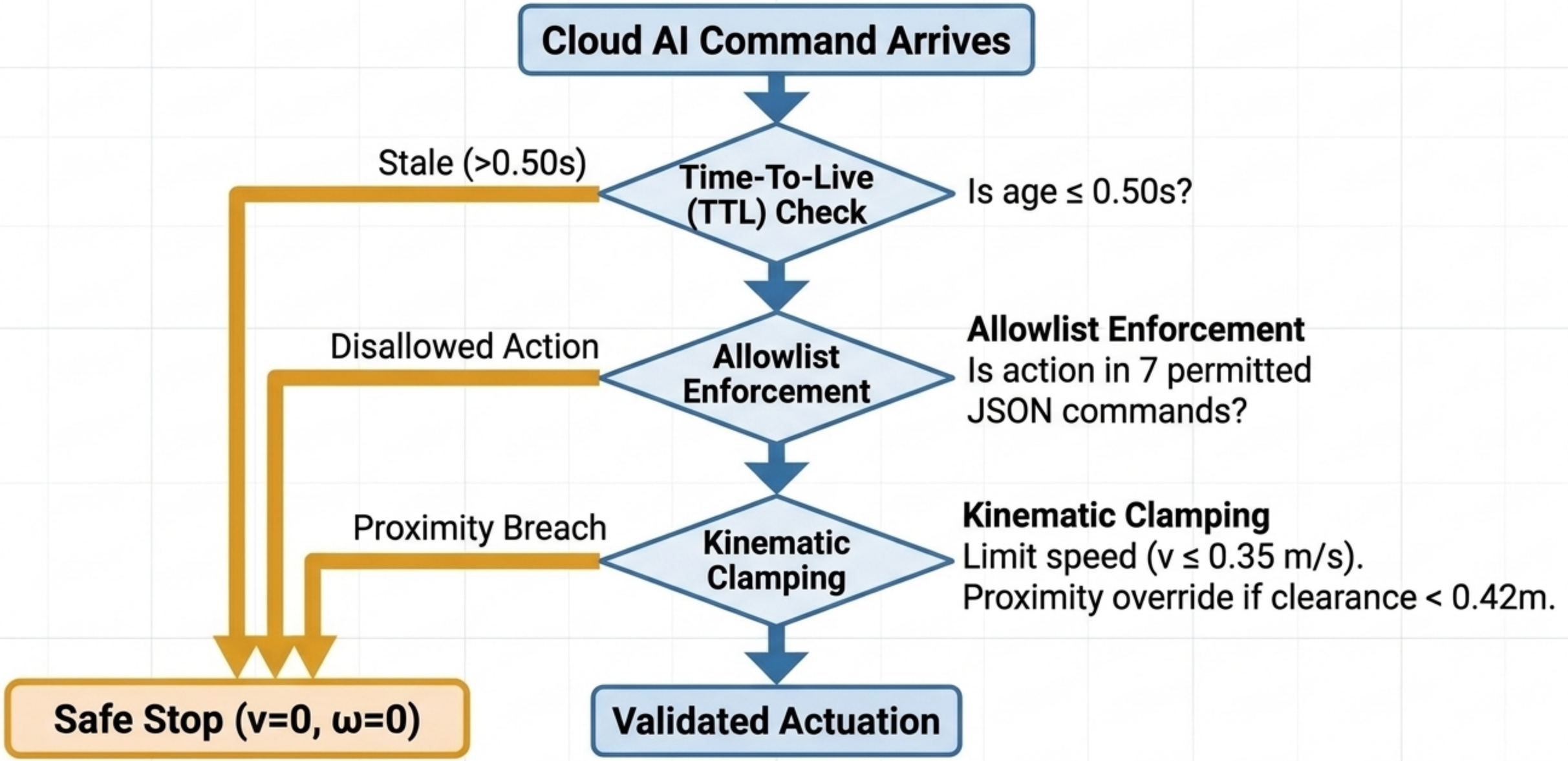
# Sensor-Integrity Monitor (SIM) Detection




**Trigger:** When any node fails, SIM packages the anomaly breakdown and sends it to the Nemotron-3-super-120b agent.

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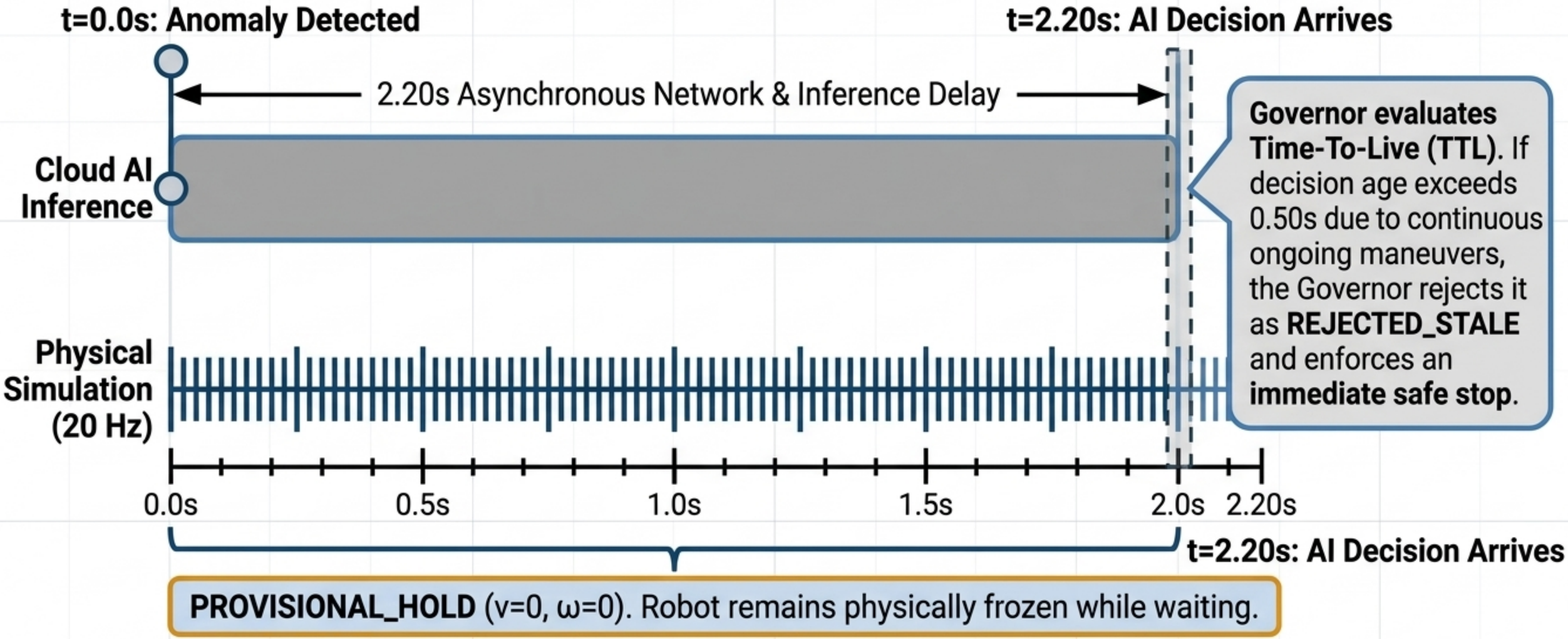
# The Independent Safety Governor (TTL & Guardrails)




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# 20 Hz Runtime Execution Sequence

Synchronous physical kinematics vs. asynchronous AI reasoning.




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# 4-Variant Comparative Evaluation Model

**Raw Telemetry**

**Variant A: Healthy Baseline**

Clean telemetry. Zero injected faults.
Establishes maximum nominal trajectory.



**Variant B: Unmitigated Controller**

Corrupted telemetry without SIM or Governor.
Follows stale/drifted data directly to physical collision.



**Variant C: Conventional Rule-Based**

Deterministic monitor triggers an unconditional, permanent halt (CONVENTIONAL_SAFE_STOP) upon the first anomaly.



**Variant D: FaultShield System**

The full stack. SIM + Nemotron LLM Agent + TTL Safety Governor.
Navigates, diagnoses, and safely recovers.



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# 5 Controlled Benchmark Experiments

| Experiment ID | Fault Type | Duration Window | Primary Failure Mode Tested |
|---|---|---|---|
| exp1_healthy | NONE | 0.0s to 0.0s (No Delay) | Zero-fault baseline navigation. |
| exp2_frozen_lidar | LIDAR_FROZEN | 2.5s to 8.0s (Delay: 0.10s) | Blind collision risk with storage racks. |
| exp3_contradictory | IMU_BIAS | 2.0s to 7.0s (Delay: 0.10s) | Cross-sensor kinematic divergence. |
| exp4_delayed_ai | AI_DECISION_DELAY | 2.0s to 10.0s (Delay: 2.20s) | Governor rejection of stale cloud AI. |
| exp5_compound | COMPOUND | 2.0s to 9.0s (Delay: 1.80s) | Simultaneous frozen LiDAR, odometry drift, IMU bias, and inference delay. |

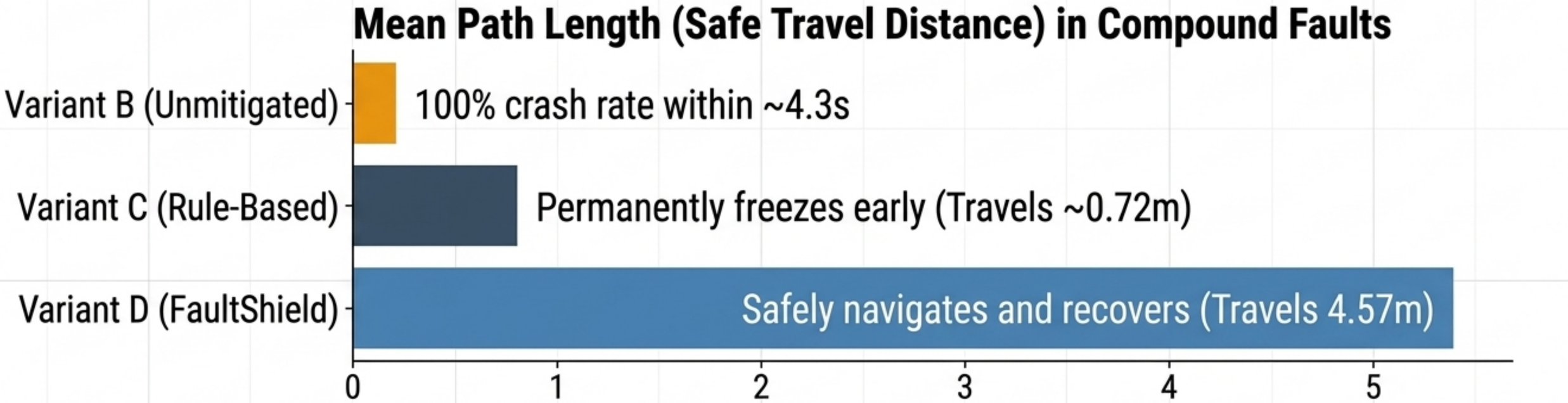
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# Quantitative Benchmark Results

**0.0%**
**Collision Rate**
(Across all scenarios)

**100%**
**Recovery Rate**
(In sensor-fault scenarios)

**0.0%**
**Stale-Response Acceptance**
(26 late decisions perfectly blocked)




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# The Architectural Core

## Why LLM recovery requires deterministic physical guardrails.




LLM-based recovery agents cannot be trusted with direct physical actuation. Real-world reliability requires pairing high-level contextual reasoning with an uncompromisable, low-level Time-To-Live (TTL) kinematic governor.

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